

Curriculum Vitae for Pedro Gomes Pinto

Master's Student in Fundamental Mathematics

Institut Élie Cartan de Lorraine · UFR MIM

Université de Lorraine, Metz

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Education

Université de Lorraine, Metz

2023–present

Master of Science, Mathématiques Fondamentales et Appliquées (research orientation). Coursework in differential geometry, algebraic topology, analysis, algebra and probability. Expected completion 2027.

Université de Lorraine, Metz

2023

Licence (B.Sc.), Mathematics and Applications.

Lycée Fabert, Metz

2020–2022

Classes Préparatoires aux Grandes Écoles (CPGE), intensive preparatory program in mathematics.

Research Interests

Differential geometry and algebraic topology, with a focus on:

- **Lorentzian geometry.** Global structure of Lorentzian spacetimes; causality theory, including global hyperbolicity and chronology protection; the geometry of spacelike foliations and the initial value problem in general relativity.
- **Spin geometry and orientability.** Orientability of smooth manifolds and its physical consequences (chirality, spin structures); Clifford algebras, spinor bundles and the Dirac operator.
- **Spectral geometry.** Eigenvalue estimates for the Laplace–Beltrami operator and Reilly-type inequalities; the influence of curvature, volume and boundary conditions on spectral bounds.
- **Topological methods in mathematical physics.** Fiber bundles and foliations over parameter spaces; superspace and the space of 3-geometries; category- and sheaf-theoretic formalizations of inter-theoretic relations.

Research Projects and Theses

Reilly-Type Inequalities in a Variety of Ambient Manifolds

Apr.–July 2026

Master's memoir (mémoire de M2), written during the end-of-degree research internship (stage de recherche), Université de Lorraine. Supervisor: Jean-François Grosjean. Reilly-type inequalities and the resulting bounds on the first eigenvalue of the Laplace–Beltrami operator across different classes of ambient manifolds, with attention to how geometric constraints (curvature, volume, boundary conditions) govern the estimates.

La cohomologie de De Rham et l'orientabilité des variétés lisses

2025

Research memoir, Université de Lorraine. Supervisor: Hervé Oyono-Oyono. De Rham cohomology and the orientability of smooth manifolds.

Application de la théorie des nœuds à l'ADN 2023
Bachelor's thesis, Université de Lorraine. Supervisor: Camille Laurent-Gengoux. Collaborative project. Topological invariants of knots, with applications to the topology of DNA.

Independent Reading Projects

Algèbres de Clifford, spineurs et l'opérateur de Dirac 2026
Self-directed reading project (in French, approx. 32 pp.). An attempt to assemble a unified, self-contained account of Clifford algebras, spinor representations and the Dirac operator from treatments scattered across the standard literature, working towards a global view of the subject rather than a summary of any single source. Includes appendices on the algebraic prerequisites and on characteristic classes (w_1 , w_2), and a section on applications in physics (Pauli spinors, the Dirac equation, Seiberg–Witten invariants). Principal sources: Lawson–Michelsohn, *Spin Geometry*, and Bourguignon, Hijazi, Milhorat, Moroianu and Moroianu, *A Spinorial Approach to Riemannian and Conformal Geometry*; conventions follow Lawson–Michelsohn.

Schools, Workshops, and Seminars Attended

SCALPOTMAS 2026 — Masterclass on Scalar Curvature and the Positive Mass Theorem 2026
Seminar on Lie Algebroids and Atiyah Obstruction Classes May 2026
Lecture series given by Panagiotis Batakidis, UFR MIM, Université de Lorraine, Metz.

Teaching and Outreach

To be completed as teaching duties are undertaken.

Other Employment

Carrefour City, Montigny-lès-Metz 2022–2024
Part-time sales associate. Inventory management and replenishment, point-of-sale operation, and digital content creation for the store.

Technical Skills

- **Typesetting:** L^AT_EX (advanced; including TikZ and preparation of long structured documents), Markdown.
- **Programming and tools:** Python, Git and GitHub, Android Studio.
- **Software:** *ResearchLibrary* — a personal Android application for organising and consulting mathematical references, designed, built and deployed independently (Android Studio; source at github.com/gppedro3714/ResearchLibrary).

Languages

- Portuguese — native
- French — fluent
- English — fluent
- German — intermediate
- Spanish — intermediate

- Luxembourgish — beginner
- Italian — beginner

Last updated: July 2026.